PATRICK FROILAN S. DEL AYRE

Full-Stack Web, Native Android & Unity 2D Game Developer

Calumpit, Bulacan, Philippines | patrickdelayre30@gmail.com | github.com/Natoy0123

PROFESSIONAL SUMMARY

Senior BS Information Technology student (BSIT 4A-G1) at Bulacan State University specializing in end-to-end software engineering across React 19 web platforms, native Java Android applications, Node.js/Express REST APIs, real-time geospatial telemetry (Leaflet/MapLibre), and 120 FPS Unity 6 2D mobile games.

TECHNICAL SKILLS

Languages: Java (Android SDK), JavaScript (ES6+), C# (Unity 6), HTML5, CSS3, SQL
Frontend & GIS: React 19, Vite, Tailwind CSS, Leaflet.js, MapLibre SDK, GeoJSON, Responsive WCAG UI
Backend & Data: Node.js, Express 5, MongoDB, Mongoose, Server-Sent Events (SSE), JWT & OTP Auth, REST Bridge APIs
Mobile & QA: Android Studio, CameraX, ZXing QR, Material Design 3, Unity IL2CPP ARM64, Git, Playwright E2E

FEATURED ENGINEERING PROJECTS

YTO Express - Admin & Dispatch Web Portal (Capstone Project) 2024 - 2026

React 19, Vite, Tailwind CSS, Node.js, Express, MongoDB, Leaflet GIS, SSE | github.com/Natoy0123/yto-express-web

- Engineered the central logistics command dashboard with real-time KPI progress cards, rider rankings, and PDF/CSV/XLSX exports.
- Built a 10-hub Bulacan regional geofence monitor with Satellite/Geofence/Heatmap layers and live rider GPS boundary alerts.
- Implemented bidirectional Bridge API sync (syncParcel, sendStatus, YTOQR1 enterprise QR payloads) with SSE live streaming.

YTO Express - Courier, Seller & Tracking Mobile App (Capstone Project) 2024 - 2026

Java, Android SDK, MapLibre SDK, CameraX, ZXing QR, Material Design 3 | github.com/Natoy0123/YTO-Express-App

- Developed a multi-role native Android app unifying Seller 4-step pickup booking, Customer live GPS tracking, and Rider dispatch.
- Integrated CameraX + ZXing high-speed waybill QR scanning and MapLibre turn-by-turn route polylines across Bulacan hubs.
- Enforced 10m/20m doorstep GPS geofencing before unlocking Proof of Delivery (POD) photo capture and digital e-receipts.

Hagonoy GroundWatch - Civic Telemetry & Subsidence GIS Platform 2026

React 19, Vite, Tailwind CSS v4, Leaflet.js, GeoJSON, Playwright E2E | hagonoy-groundwatch-tik7.onrender.com

- Built an open-access civic GIS platform modeling coastal land subsidence and tidal flooding across 26 Hagonoy barangays.
- Created an interactive Leaflet hazard map with Critical/Severe/Moderate polygon filtering and an Aquifer Stress Simulator.
- Implemented a citizen science incident reporting intake system with tracking receipts (HGW-XXXX) and Playwright E2E tests.

Slime Food Drop - 120 FPS Unity 2D Mobile Arcade Game 2026

Unity 6 (6000.3.0f1), C#, Android SDK, IL2CPP ARM64, 2D Physics | github.com/Natoy0123/SlimeFoodDrop

- Engineered a portrait 2D mobile arcade game featuring squash-and-stretch character physics and 10 handcrafted themed stages.
- Optimized touch input and rendering pipeline for 120 FPS mobile displays using Unity IL2CPP ARM64 native compilation.

EDUCATION & ACADEMIC MILESTONES

Bachelor of Science in Information Technology (BSIT 4A-G1) 2023 - Present (Senior)

Bulacan State University (BulSU) - Malolos, Bulacan, Philippines

- Lead Full-Stack Web & Native Android Developer for the 2-year YTO Express Logistics Capstone System (2024-2026).
- Relevant Coursework: Web Systems & Technologies, Mobile Application Development, Database Management, GIS Telemetry, QA.